

# Lab 11: SPI

Today we will work with SPI.

## Task – SPI Master Module

For those who have not already worked with this, build a generic SPI module based on the provided FSM template . How do you make the SPI module generic, such that it complies with the SPI protocol and can be used in different circuits without modification? Consider a way to make the interface between the SPI module and the remaining circuit simple and effective. Test the module in simulation and verify that it functions as expected.

## Task – FSM Controller for the SPI Module

Build a FSM which uses the SPI module to send/receive data. Utilize the simple interface you have developed for control of the SPI module from your FSM. Take inspiration from either your semester project or the assignment. Remember to verify the entity independently in simulation as well as with the SPI module – and consider making a test bench.